

Lab Wireshark

Entorno

Ubuntu

Wireshark

Virtual Box

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1. Filtro ARP

Al aplicar el filtro ARP, muestra solo dicho tráfico (requests y replies). Sirve para realizar búsquedas más limpias y eliminar todo lo que no sea ARP. Se puede detectar la MAC Address del atacante y su host:

- MAC Address: bc:24:11:52:16:9a
- Host: ProxmoxServe 52:16:9a

Wireshark interface showing a packet capture filtered by 'arp'. The packet list displays several ARP requests from ProxmoxServe_4a:e6:df to the broadcast address (ff:ff:ff:ff:ff:ff). The packet details for Frame 1 show the Ethernet II header and the ARP request structure.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
2	1.002952	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
9	2.030974	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
10	4.097450	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
14	5.098620	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
27	6.122556	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
41	10.243235	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
43	11.274258	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
44	12.302232	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
46	15.876591	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
47	16.909899	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
48	17.929836	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.11? Tell 192.168.127.5
51	19.387008	HewlettPacka_cc:b2:df	Broadcast	ARP	60	Who has 192.168.127.8? Tell 192.168.127.9
52	19.395888	ProxmoxServe_ea:20:df	Broadcast	ARP	60	Who has 192.168.127.9? Tell 192.168.127.6
53	19.397162	HewlettPacka_cc:b2:df	Broadcast	ARP	60	Who has 192.168.127.6? Tell 192.168.127.9
54	19.405966	ProxmoxServe_4a:e6:df	Broadcast	ARP	60	Who has 192.168.127.9? Tell 192.168.127.5
55	19.407482	HewlettPacka_cc:b2:df	Broadcast	ARP	60	Who has 192.168.127.5? Tell 192.168.127.9

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{33D71B1D-8C14-47B2-A948-94DB574F61DC}, id 0
Ethernet II, Src: ProxmoxServe_4a:e6:df (bc:24:11:4a:e6:df), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Address Resolution Protocol (request)

0000 ff ff ff ff ff bc 24 11 4a e6 df 0
0010 08 00 06 04 00 01 bc 24 11 4a e6 df c
0020 00 00 00 00 00 00 c0 a8 7f 0b 00 00 0
0030 00 00 00 00 00 00 00 00 00 00 00 0

Address Resolution Protocol: Protocol Paquetes: 360 - Displayed: 176 (48.9%) Perfil: Default

Con el filtro arp.duplicate-address-detected, se puede visualizar como dos hosts reclaman la misma IP. Los ataques de tipo spoof suelen duplicar las IPs

The screenshot shows the Wireshark interface with the filter 'arp.duplicate-address-detected' applied. The packet list displays several ARP requests from various sources to the destination 192.168.127.9. The packet details pane highlights the 'Duplicate IP address detected for 192.168.127.9 (bc:24:11:52:16:9a) - also in use by 38:21:c7:cc:b2:c9 (frame 55)'.

No.	Time	Source	Destination	Protocol	Length	Info
56	19.416132	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.9 is at bc:24:11:52:16:9a
70	19.474483	Fortinet_fc:14:18	Broadcast	ARP	60	Who has 192.168.127.3? Tell 192.168.127.8 (duplicate use of 192.168.127.8 detected!)
71	19.474500	ProxmoxServe_ba:52:...	Fortinet_fc:14:18	ARP	42	192.168.127.3 is at bc:24:11:ba:52:0e (duplicate use of 192.168.127.8 detected!)
77	19.518529	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.6 is at bc:24:11:52:16:9a
80	19.569606	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.5 is at bc:24:11:52:16:9a
83	19.590116	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.9 is at bc:24:11:52:16:9a
92	19.610501	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.6 is at bc:24:11:52:16:9a
94	19.620715	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.5 is at bc:24:11:52:16:9a
98	19.630820	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.1 is at bc:24:11:52:16:9a
102	19.681969	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.1 is at bc:24:11:52:16:9a
107	20.722935	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.9 is at bc:24:11:52:16:9a
109	20.824902	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.6 is at bc:24:11:52:16:9a
110	20.875805	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.5 is at bc:24:11:52:16:9a
111	20.896117	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.9 is at bc:24:11:52:16:9a
113	20.916541	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.6 is at bc:24:11:52:16:9a
114	20.926627	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.5 is at bc:24:11:52:16:9a
115	20.936822	ProxmoxServe_52:16:...	ProxmoxServe_ba:52:...	ARP	60	192.168.127.1 is at bc:24:11:52:16:9a

Frame 56: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{33071B1D-8C14-47B2-A948-94DB574F61DC}, id 0
Ethernet II, Src: ProxmoxServe_52:16:9a (bc:24:11:52:16:9a), Dst: ProxmoxServe_ba:52:0e (bc:24:11:ba:52:0e)
Address Resolution Protocol (reply)
Duplicate IP address detected for 192.168.127.9 (bc:24:11:52:16:9a) - also in use by 38:21:c7:cc:b2:c9 (frame 55)

Paquetes: 360 - Displayed: 64 (17.8%)

2. Filtro ICMP

El filtro icmp muestra las solicitudes y respuestas de ping, lo que hace más fácil inspeccionar TTL y descubrir el SO del atacante

The screenshot shows the Wireshark interface with the filter 'icmp' applied. The packet list displays several ICMP Echo (ping) requests from source 8.8.8.8 to destination 192.168.127.3. The packet details pane highlights the 'Internet Control Message Protocol'.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58169/14819, ttl=64 (no response found!)
2	0.000000	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58425/14820, ttl=64 (no response found!)
3	0.000000	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58681/14821, ttl=64 (no response found!)
4	0.000024	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58937/14822, ttl=64 (no response found!)
5	0.000029	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59193/14823, ttl=64 (no response found!)
6	0.000052	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59449/14824, ttl=64 (no response found!)
7	0.000072	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59705/14825, ttl=64 (no response found!)
8	0.000072	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59961/14826, ttl=64 (no response found!)
9	0.000094	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60217/14827, ttl=64 (no response found!)
10	0.000118	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60473/14828, ttl=64 (no response found!)
11	0.000118	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60729/14829, ttl=64 (no response found!)
12	0.000146	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60985/14830, ttl=64 (no response found!)
13	0.000151	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=61241/14831, ttl=64 (no response found!)
14	0.000178	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=61497/14832, ttl=64 (no response found!)
15	0.000178	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=61753/14833, ttl=64 (no response found!)
16	0.000206	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=62009/14834, ttl=64 (no response found!)
17	0.000222	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=62265/14835, ttl=64 (no response found!)

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{33071B1D-8C14-47B2-A948-94DB574F61DC}, id 0
Ethernet II, Src: ProxmoxServe_52:16:9a (bc:24:11:52:16:9a), Dst: ProxmoxServe_ba:52:0e (bc:24:11:ba:52:0e)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.127.3
Internet Control Message Protocol

Paquetes: 210902 - Displayed: 210902 (100.0%)

3. Filtro IP.TTL

Aplicando `ip.ttl == 64` se puede buscar directamente sobre el TTL específico (en este caso 64). Los TTL asignados son:

- 64 = Linux/Unix
- 128 = Windows
- 255 = Cisco

Wireshark capture showing ICMP Echo (ping) requests from 8.8.8.8 to 192.168.127.3. The filter is `ip.ttl == 64`. The packet list shows 17 entries, all with TTL=64 and 'no response found!'. The packet details pane shows the selected packet as an Internet Control Message Protocol (ICMP) Echo (ping) request.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58169/14819, ttl=64 (no response found!)
2	0.000000	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58425/14820, ttl=64 (no response found!)
3	0.000000	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58681/14821, ttl=64 (no response found!)
4	0.000024	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=58937/14822, ttl=64 (no response found!)
5	0.000029	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59193/14823, ttl=64 (no response found!)
6	0.000052	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59449/14824, ttl=64 (no response found!)
7	0.000072	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59705/14825, ttl=64 (no response found!)
8	0.000072	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=59961/14826, ttl=64 (no response found!)
9	0.000094	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60217/14827, ttl=64 (no response found!)
10	0.000118	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60473/14828, ttl=64 (no response found!)
11	0.000118	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60729/14829, ttl=64 (no response found!)
12	0.000146	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=60985/14830, ttl=64 (no response found!)
13	0.000151	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=61241/14831, ttl=64 (no response found!)
14	0.000178	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=61497/14832, ttl=64 (no response found!)
15	0.000178	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=61753/14833, ttl=64 (no response found!)
16	0.000206	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=62009/14834, ttl=64 (no response found!)
17	0.000222	8.8.8.8	192.168.127.3	ICMP	60	Echo (ping) request id=0x1584, seq=62265/14835, ttl=64 (no response found!)

Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{33D7181D-8C14-47B2-A948-94DB574F61DC}, id 0
Ethernet II, Src: ProxmoxServe_52:16:9a (bc:24:11:52:16:9a), Dst: ProxmoxServe_ba:52:0e (bc:24:11:ba:52:0e)
Internet Protocol Version 4, Src: 8.8.8.8, Dst: 192.168.127.3
Internet Control Message Protocol

4. Resumen final

Se detectó un ataque de IP spoofing (suplantación de la IP 8.8.8.8) mediante ARP spoofing, donde se observaron IPs y MACs inconsistentes. El análisis reveló que el atacante usa una MAC real con TTL típico de Linux/Unix y una interfaz virtual de Proxmox.